

João Marcos Ferreira Vilela

Full-Stack Engineer — AI Integrations, Messaging Automation & Production Operations

Caruaru, Brazil (UTC-3) · Open to remote — full US/EU morning overlap

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English — C1 Advanced (Duolingo English Test 130, 2026) · Portuguese — Native

SUMMARY

I build the integrations that connect business systems to WhatsApp, the automated support that runs on top of them, and the operations layer that keeps both alive. Currently shipping ERP ↔ WhatsApp automation across the vehicle-protection and accounting sectors. Oracle-certified in Generative AI, and I use an LLM agent workflow as part of how I develop. I own what happens after deploy — replication monitoring, backup verification, health checks — not just application code.

EXPERIENCE

Technical Operations Analyst · Átomo Soluções e Gestão

Mar 2026 – Present

Full-time · Caruaru, Brazil — B2B services & automation

- **Chatbot platform lead — 23 associations running WhatsApp bots in production**, 14 of them onboarded within a single 4-month window after I standardized the base flow, turning client onboarding from a bespoke project into a repeatable process. 73 of 102 platform tasks in the period; 32 were payment and API integrations.
- **Amora — AI agent in production** (*built with one other engineer*): answers accounting and tax questions for a client's members and **escalates to the right human team on its own**. Prompts versioned in Git across 6 documented refinement cycles.
- **Self-service billing over WhatsApp**: members request a boleto, a duplicate invoice or a PIX and the bot pulls it from the ERP and delivers it — 28 tasks across build, automation and fixes, the platform's most recurring flow.
- **Dispatch system for a 24h roadside-assistance company (sole developer)**: LNSoft polling, Supabase, PM2, batching, blocklists, LGPD handling and a resilient cron. Built Apr–Jun 2026, in production.
- **ERP ↔ WhatsApp integrations** across 6 platforms — Hinova SGA, South, DevSul, EPTA, Conta Azul and Asaas — automating billing, boleto/PIX delivery and invoice notices. (*Node.js, TypeScript, Hono, Supabase*)
- **Reliability engineering**: Postgres replication monitor (lag, apply delay, retained WAL, offline-slot alerting), 2-hourly backups with 24-copy rotation and S3 mirroring, and a target-verification check that aborts a backup pointing at the wrong destination. Operate a financial ERP (.NET, PostgreSQL/TimescaleDB) on Docker Swarm + Traefik and a customer-facing app in PM2 cluster mode.
- **126 verified technical deliveries in 4 months** (Mar–Jul 2026), 47 of them outside the ticket system, documented by Git history and daily check-outs — counted by project-day, never by commit.

Co-Founder & Full-Stack Engineer · Gnomon

Oct 2025 – Jun 2026

Indoor-navigation startup · Caruaru, Brazil

- Sole developer of the system core: routing algorithms, API and React UI for an indoor-wayfinding PWA for campuses, hospitals and event venues.
- Full-TypeScript architecture front to back, with end-to-end coverage in Cypress.
- Ranked **1st among all Systems Analysis cohorts** and advanced through **Centelha 3**, a Brazilian government innovation program.

Freelance Web Developer

2023 – 2025

Independent

- Delivered production sites for paying clients: **SolarTech** (energy-savings platform with simulation dashboards, React + Vite) and **Renova Aesthetic** (e-commerce for an aesthetics clinic, Next.js + Stripe).

SELECTED PROJECTS

Personal systems I run in production, plus clean-room rebuilds of patterns I shipped at work. Live demos at joaovilela.vercel.app.

Touvie — Personal life OS · Next.js 15, Supabase, Telegram, PWA
An AI assistant that **writes to real data** through tool calls — creating goals, logging transactions — not just answering questions. Device-trust middleware on signed HMAC-SHA256 cookies; Supabase row-level security proven by E2E tests asserting one user cannot read another's records. 320+ commits, 40+ migrations, Playwright suite.

Caçador de Ofertas — Deal-hunting bot · Next.js, Supabase, Telegram
Parses price, coupon and store out of free-form Portuguese across 25 Telegram channels every 5 minutes. Separating coupon values from product prices is what made the alerts trustworthy — read naively, "R\$30 OFF" becomes a R\$30 product and satisfies every hunt. Ships a dead-man's-switch canary that catches a silently broken scraper. **470+ passing tests**.

allchats — Multi-tenant WhatsApp support bot · Next.js 16, React 19, Claude SDK
RAG-grounded answers with guardrails and human escalation via tool use. Runs correctly with **no API key configured** — CI exercises that deterministic fallback path on purpose, so a provider outage degrades instead of breaking.

Also: **cobraflow** (WhatsApp collections dispatch — SSE batch engine, two-layer blocklist, audit log) · **agendapp** (booking PWA, concurrency-tested so two people can never win the same slot) · **Zaptutor** (Chrome MV3 extension, zero network calls).

TECHNICAL SKILLS

Languages	TypeScript · JavaScript · Python · SQL · C#
AI Engineering	LLM tool-calling (Claude, GPT) · RAG with guardrails · prompt engineering · agent-assisted development workflow
Back-end	Node.js (Hono, Express) · FastAPI · .NET · REST API design · Server-Sent Events
Front-end	React · Next.js (App Router, SSR) · Tailwind CSS
Data	PostgreSQL · TimescaleDB · Supabase (RLS) · Prisma · Alembic
Integrations	WhatsApp Business & Cloud API (Meta templates) · ERP APIs · Google Calendar API · Chatwoot · n8n
Infrastructure	Docker Swarm + Traefik · PM2 cluster mode · systemd · AWS (S3, EC2) · Oracle Cloud · Vercel · Fly.io
Reliability	replication monitoring · backup rotation & verification · health-check endpoints · dry-run and blocklist patterns for irreversible actions
Testing	Playwright · Cypress · Vitest · concurrency and cross-tenant isolation testing

EDUCATION & CERTIFICATIONS

Associate Degree, Systems Analysis and Development — UNINASSAU, Brazil Jan 2024 – Dec 2025
GPA 8.6/10 · Perfect scores in Software Architecture, Cloud Computing and Machine Learning.

- **Oracle Certified Generative AI Professional** — Oracle, Sep 2025
- **CS50: Introduction to Computer Science** — Harvard, 2026 (*in progress*)
- **Python Back-End Bootcamp** — DIO & Santander, 2025 (58h)
- **AWS Educate: Introduction to Cloud 101** — AWS, Jul 2025